



UNIVERSIDAD TECNOLÓGICA DE PANAMÁ

FACULTAD DE INGENIERÍA DE SISTEMAS COMPUTACIONALES

LICENCIATURA EN CIBERSEGURIDAD

Ejercicio Practico 1

Asignatura:

Programación II

Integrantes:

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Profesor:

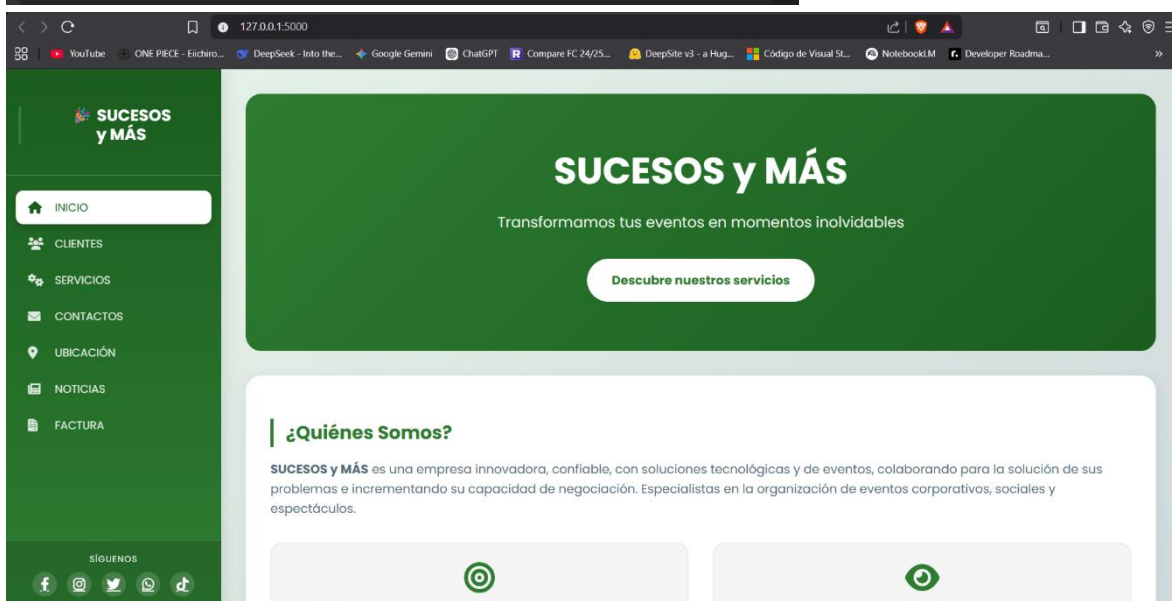
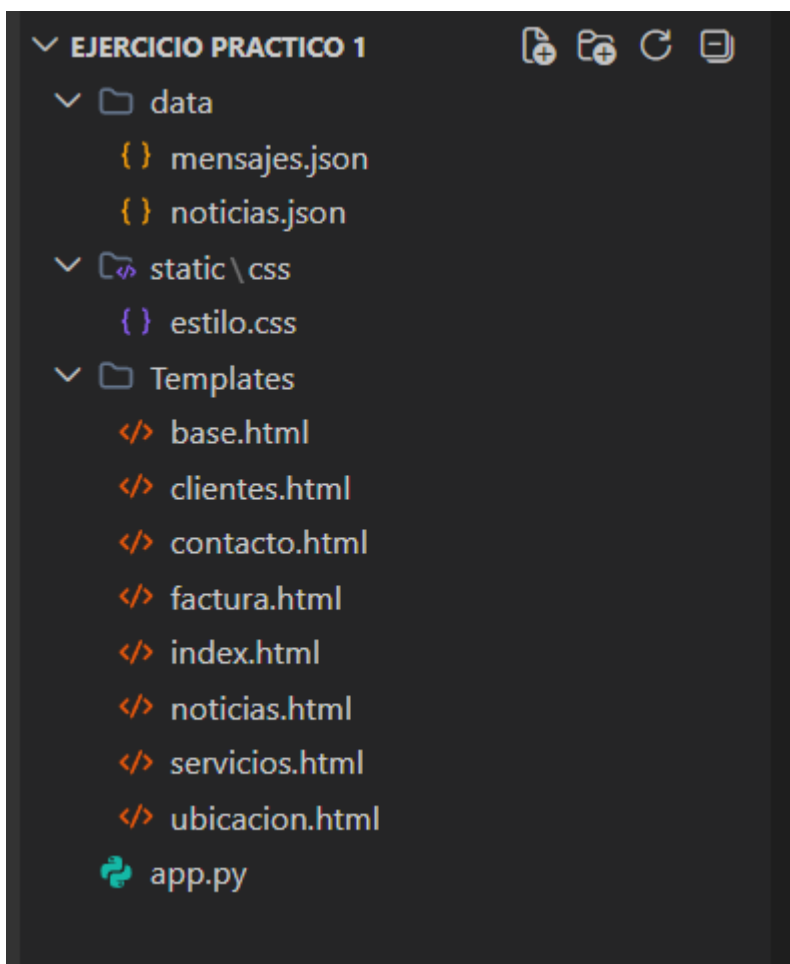
Napoleon Ibarra

Semestre:

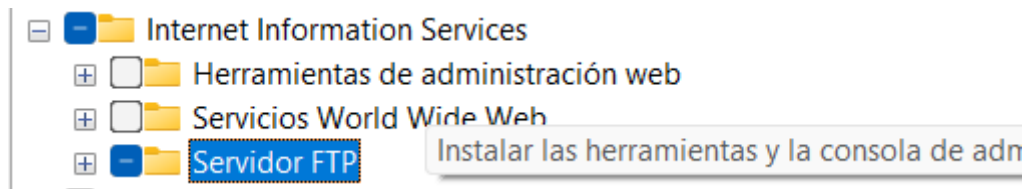
Primer Semestre

2026

Parte 1. Creacion de la pagina WEB utilizando flask.



Parte 2. Estableciendo una conexión por FTP desde Windows a una VM (Ubuntu) y también SSH.



```
adrianprog@adrian-progII:~$ sudo apt update
Obj:1 http://archive.ubuntu.com/ubuntu noble InRelease
Obj:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Obj:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Obj:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Se puede actualizar 1 paquete. Ejecute «apt list --upgradable» para verlo.
adrianprog@adrian-progII:~$ sudo apt install vsftpd -y
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Se instalarán los siguientes paquetes NUEVOS:
  vsftpd
0 actualizados, 1 nuevos se instalarán, 0 para eliminar y 1 no actualizados.
Se necesita descargar 120 kB de archivos.
Se utilizarán 312 kB de espacio de disco adicional después de esta operación.
Des:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 vsftpd amd64 3.0.5-0ubu
ntu3.1 [120 kB]
Descargados 120 kB en 1s (196 kB/s)
Preconfigurando paquetes ...
Seleccionando el paquete vsftpd previamente no seleccionado.
(Leyendo la base de datos ... 199925 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar ../vsftpd_3.0.5-0ubuntu3.1_amd64.deb ...
Desempaquetando vsftpd (3.0.5-0ubuntu3.1) ...
Configurando vsftpd (3.0.5-0ubuntu3.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/vsftpd.service → /usr/lib/s
ystemd/system/vsftpd.service.
Procesando disparadores para man-db (2.12.0-4build2) ...
adrianprog@adrian-progII:~$
```

```
adrianprog@adrian-progII:~$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-28 04:23:12 EST; 1min 18s ago
     Process: 27657 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, stat>
   Main PID: 27659 (vsftpd)
      Tasks: 1 (limit: 4597)
     Memory: 700.0K (peak: 1.2M)
        CPU: 16ms
      CGroup: /system.slice/vsftpd.service
              └─27659 /usr/sbin/vsftpd /etc/vsftpd.conf

abr 28 04:23:12 adrian-progII systemd[1]: Starting vsftpd.service - vsftpd FTP server.>
abr 28 04:23:12 adrian-progII systemd[1]: Started vsftpd.service - vsftpd FTP server.
lines 1-13/13 (END)
```

```
#
# Uncomment this to enable any form of FTP write command.
write_enable=YES
#
# Default umask for local users is 077. You may wish to change this to 022,
```

```
adrianprog@adrian-progII:~$ sudo ufw allow 21/tcp
Reglas actualizadas
Reglas actualizadas (v6)
```

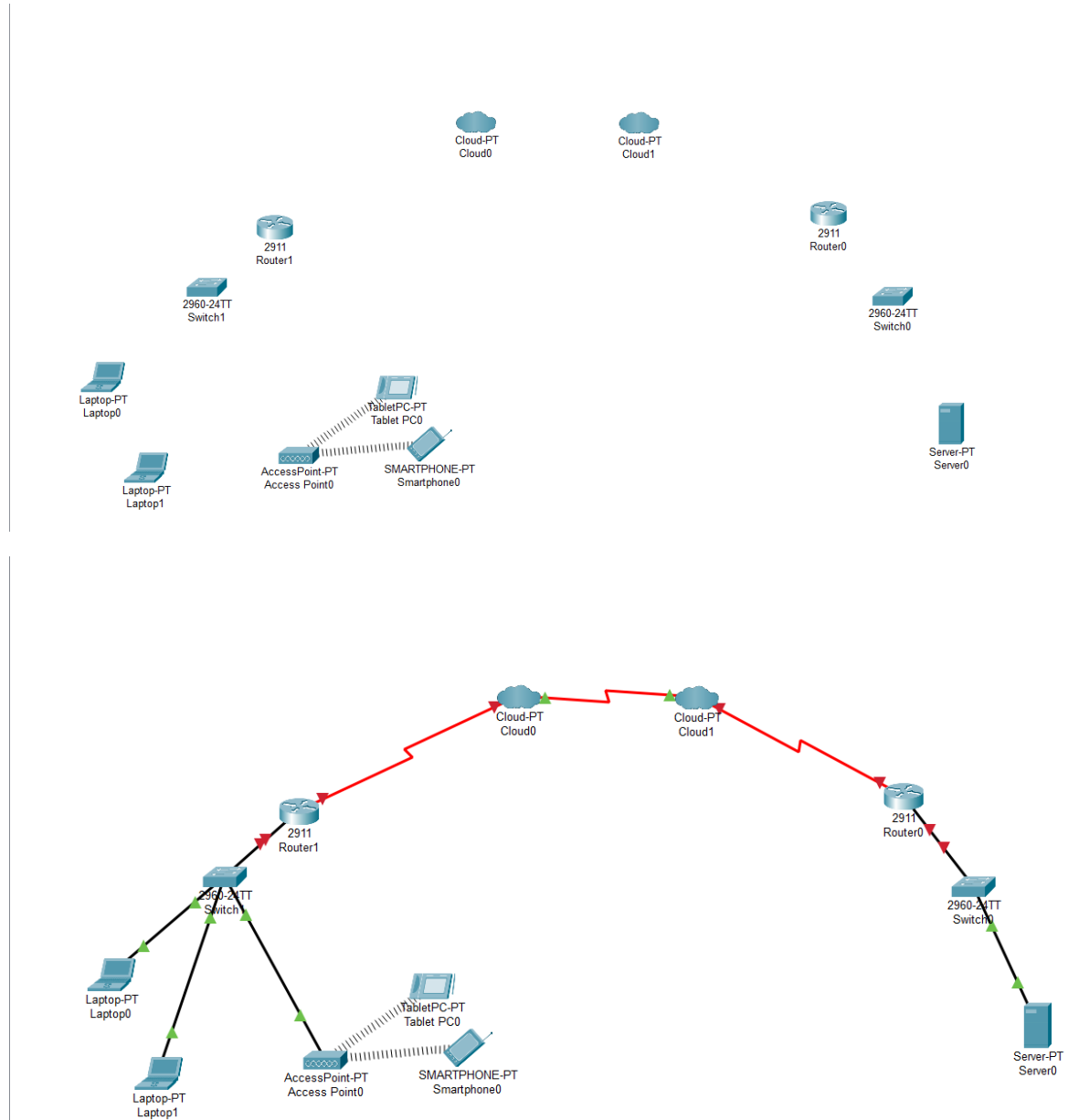
```
adrianprog@adrian-progII:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-28 03:52:03 EST; 42min ago
 TriggeredBy: ● ssh.socket
    Docs: man:sshd(8)
          man:sshd_config(5)
   Main PID: 1335 (sshd)
     Tasks: 1 (limit: 4597)
    Memory: 2.1M (peak: 2.6M)
       CPU: 135ms
    CGroup: /system.slice/ssh.service
           └─1335 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

abr 28 03:52:02 adrian-progII systemd[1]: Starting ssh.service - OpenBSD Secure Shell s>
abr 28 03:52:02 adrian-progII sshd[1335]: Server listening on 0.0.0.0 port 22.
abr 28 03:52:03 adrian-progII sshd[1335]: Server listening on :: port 22.
abr 28 03:52:03 adrian-progII systemd[1]: Started ssh.service - OpenBSD Secure Shell s>
lines 1-17/17 (END)
```

```
ftp> put "C:\Users\adria\Documents\3 Semestre\Programacion II\prueba_ubuntito.docx"
200 PORT command successful. Consider using PASV.
150 Ok to send data.
226 Transfer complete.
ftp> |
```

```
adrianprog@adrian-progII:~$ ls -R
.:
Descargas      Escritorio     Público
Documentos    Imágenes      sistema_completo.json
ejecucion.log  Música        snap
ejecucion_taller1.log Plantillas     Vídeos
ejecucion_taller1.py 'PROBELMAS 1 Y 2.py'
ejecutar_taller.sh  prueba_ubuntito.docx
```

Parte 3.Creacion de una red LAN en cisco packet tracer.



```

Router>enable
Router#interface G0/0
      ^
% Invalid input detected at '^' marker.

Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface G0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface S0/0/0
%Invalid interface type and number
Router(config)#interface S0/3/0
Router(config-if)#encapsulation frame-relay
Router(config-if)#ip address 190.0.0.1 255.255.255.0
Router(config-if)#frame-relay map ip 190.0.0.2 100 broadcast
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config-if)#

```

```

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

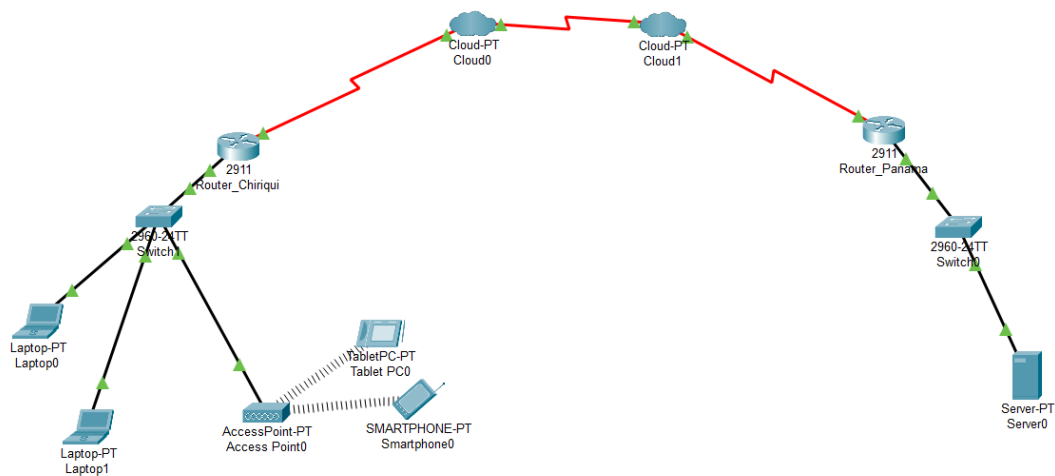
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface S0/3/0
Router(config-if)#encapsulation frame-relay
Router(config-if)#ip address 190.0.0.2 255.255.255.0
Router(config-if)#frame-relay map ip 190.0.0.1 100 broadcast
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

```



```

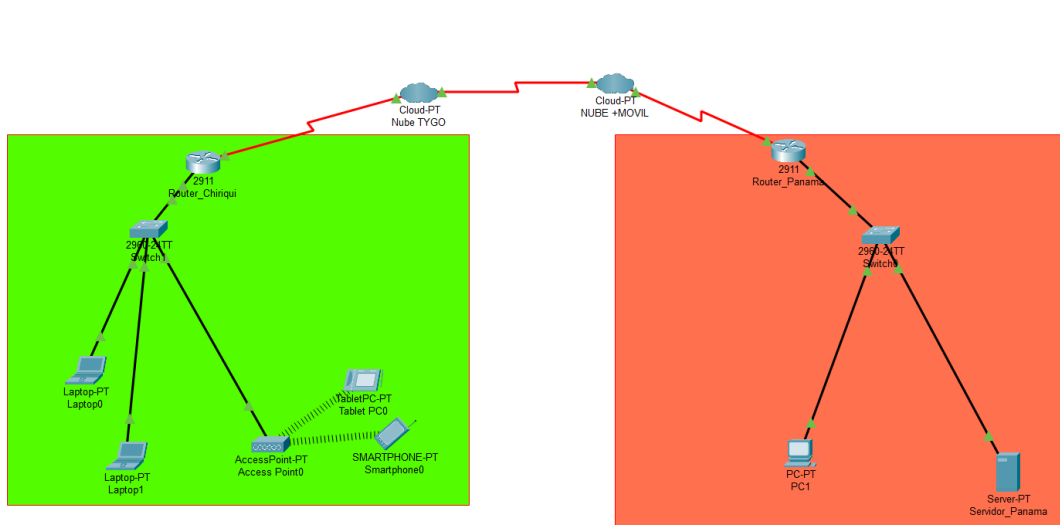
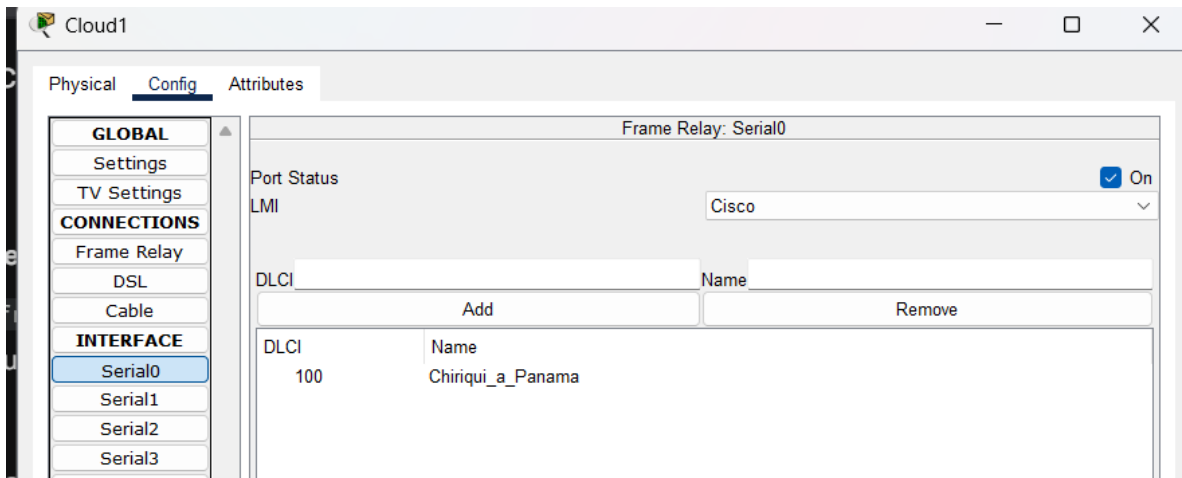
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.1.1
Router(config)#ip dhcp pool RED_CHIRIQUI
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#

```

```

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.10.1
Router(config)#ip dhcp pool RED_PANAMA
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

```



Fire	Last Status	Source	Destination	Type	Color	Time(sec)
	Successful	Laptop1	Servidor_...	ICMP		0.000